

TRANSMISSION MEDIA



Topics



Transmission media

- ☐ Definition
- ☐ Types of Transmission Media
 - ☐ Guided Media
 1. Twisted Pair cable
 2. Coaxial Cables
 3. Optical Fiber
 - ☐ Unguided media
 1. Microwave
 2. Infrared
 3. Bluetooth
 4. WI-FI (Wireless Fidelity)
- ☐ Applications

What is Transmission Media

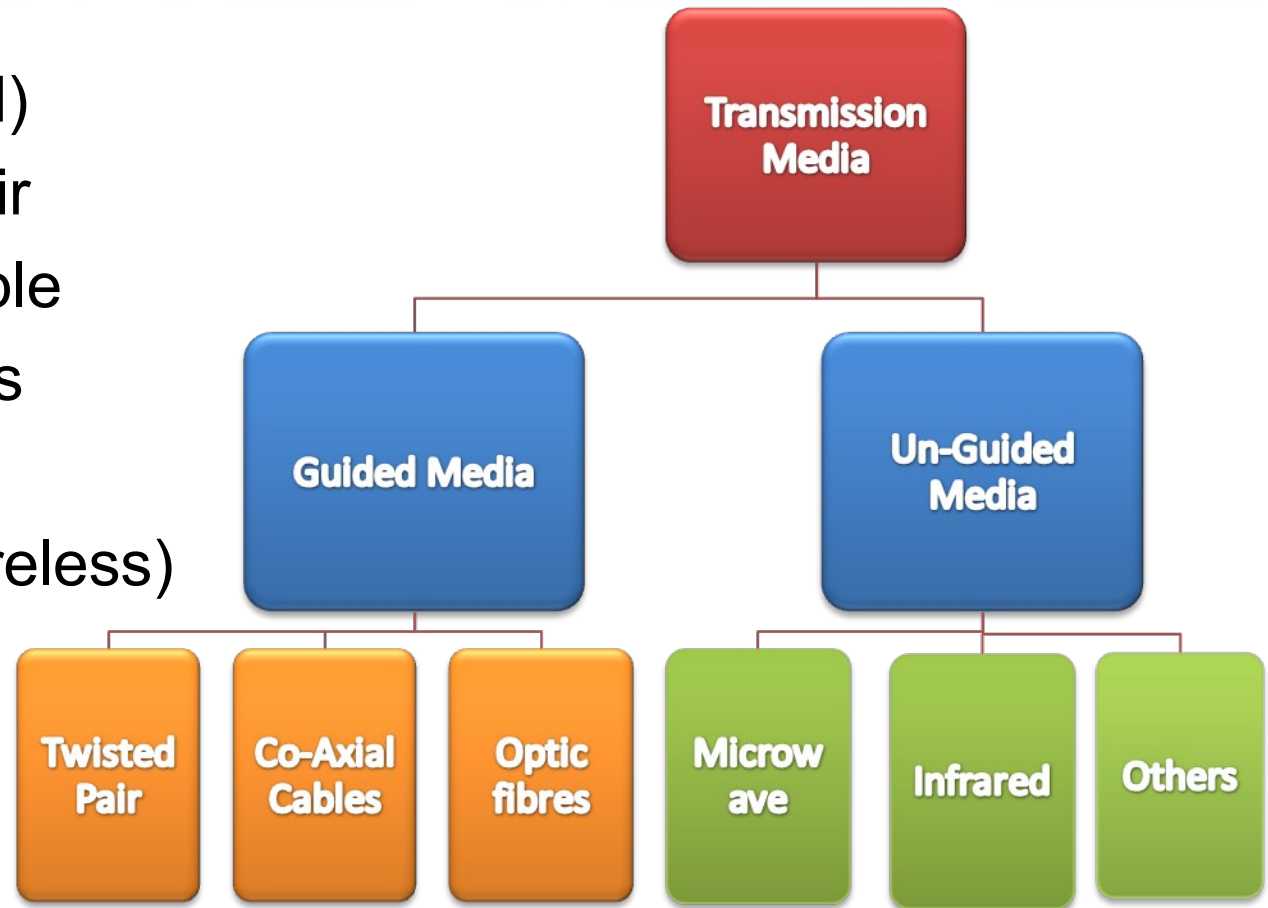


- Sending of data from one device to another is called transmission of data
- Medium used to transmit the data is called Media
- Transmission of data through Medium is called Transmission Media

Types Of Transmission Media



- Guided (Wired)
 - Twisted Pair
 - Coaxial cable
 - Fiber Optics
- Unguided (Wireless)
 - Wi-Fi
 - Bluetooth
 - Infrared
 - Microwave
 - Lasers

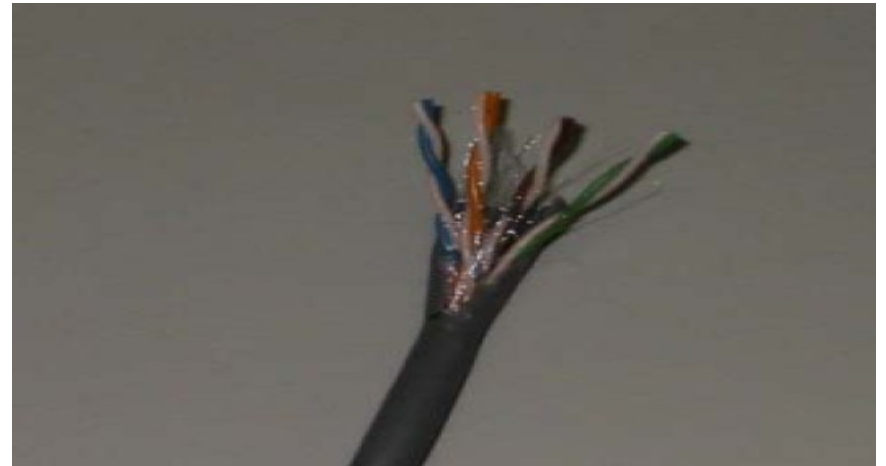
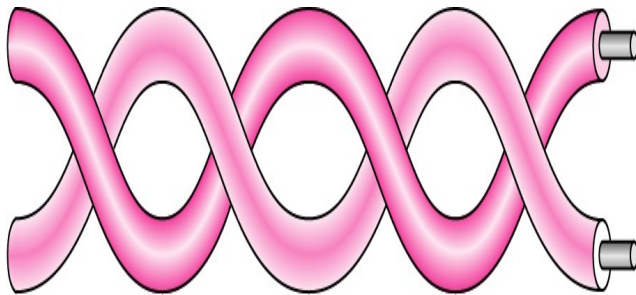


Guided (Wired)



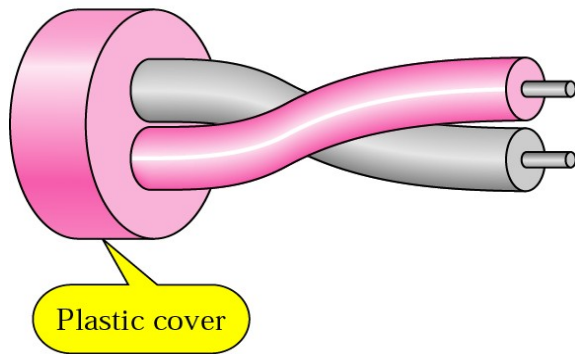
- The medium itself is more important in determining the limitations of transmission.
- The transmission capacity, in terms of either data rate or bandwidth, depends critically on the distance and on whether the medium is point-to-point or multipoint.

Twisted Pair

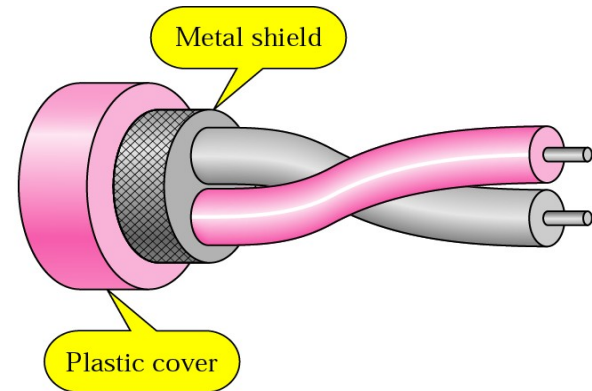


- A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.
- The twisting tends to decrease the crosstalk interference between adjacent pairs in a cable.
- Twisted pair may be used to transmit both analog and digital transmission.
- Twisted pair is limited in distance, bandwidth, and data rate.
- The attenuation for twisted pair is a very strong function of frequency

Twisted Pair Types

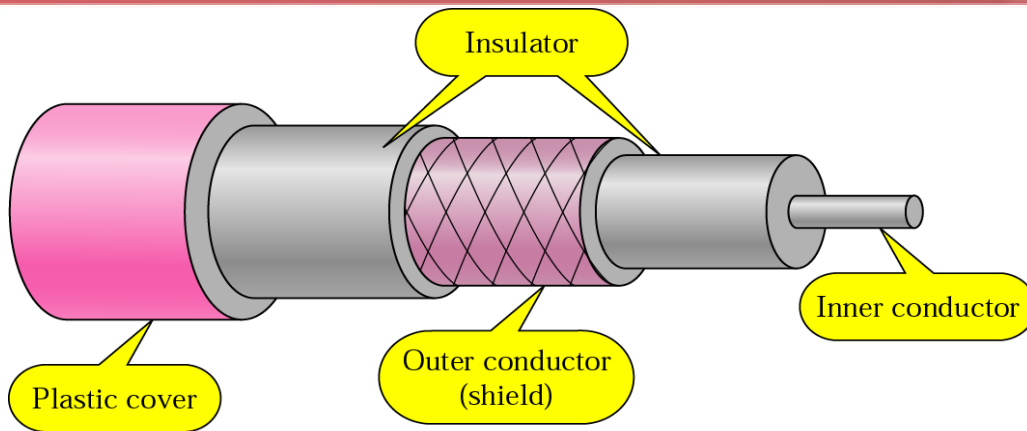


- Unshielded Twisted Pair (UTP)
 - ordinary telephone wire
 - cheapest
 - easiest to install
 - suffers from external EM interference



- Shielded Twisted Pair (STP)
 - metal braid or sheathing that reduces interference
 - more expensive
 - easiest to install
 - harder to handle (thick, heavy)

Coaxial cable

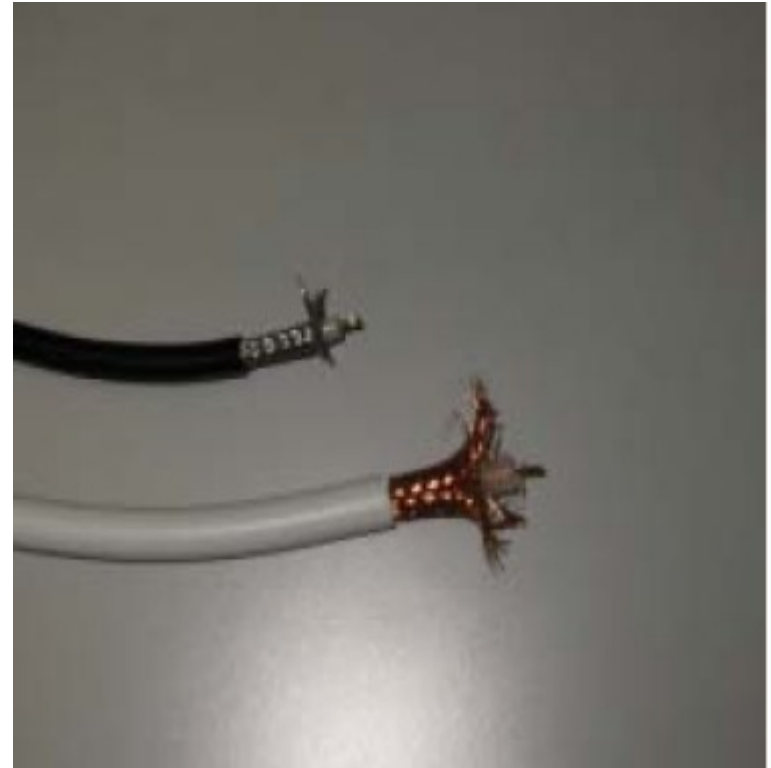


- Coaxial cable consist the followings layers in its construction
 - The copper conductor
 - Insulation layer of plastic foam
 - Second conductor or shield of wire mesh tube or metallic foil
 - Outer jacket of tough plastic
- Coaxial cable can be used over longer distances and support more stations on a shared line than twisted pair.

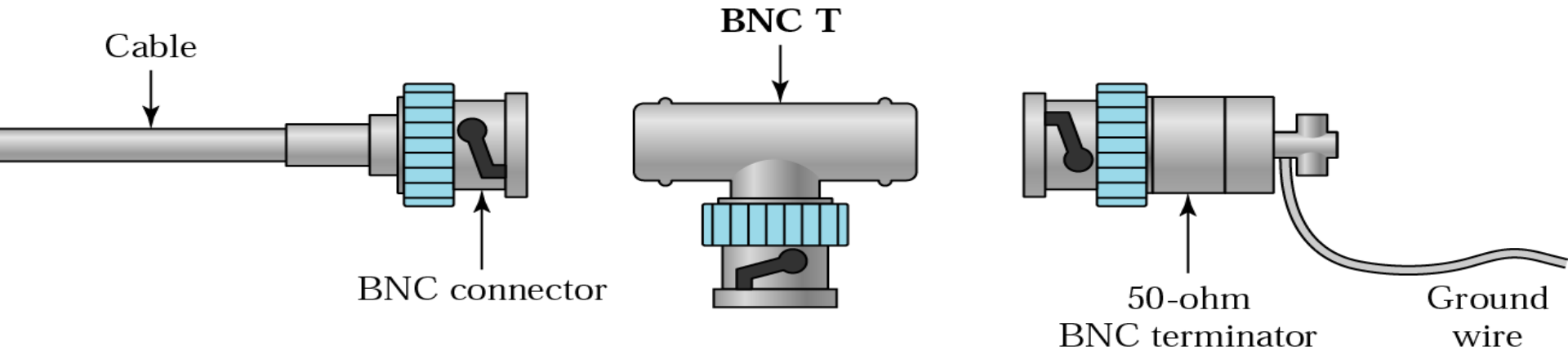
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- Coaxial cable is a versatile transmission medium, used in a wide variety of applications, including:
 - Television distribution - aerial to TV systems.
- Characteristics
 - It is comparatively inexpensive
 - Its installation is comparatively simple
 - It must be grounded properly in a network connection
 - Its bandwidth capacity is around 10 Mbps
 - It is thin Ethernet connection maximum 30 nodes and in thick Ethernet connection maximum 100 nodes can be successfully interlinked with this cable connection
 - It suffers from data attenuation

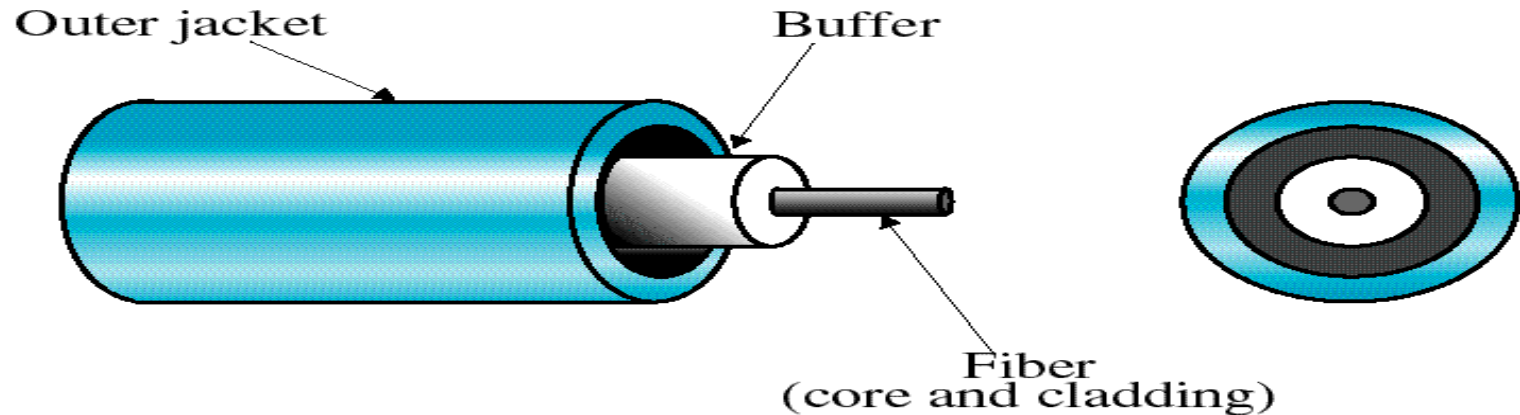


BNC connectors



- BNC Connectors are used for connecting to co-axial cables together at connecting points is when repeaters are required

Fiber Optics



- Fiber optic cable is known as most sophisticated cables used in long distance network connection.
- Through this cable data transmission is done through « Light ray signal transmission» rather than electrical signal transmission.
- It has inner core of glass that conducts light. This inner core is surrounded by cladding.
- Cladding is nothing but layer of glass material that reflects light back into the core.
- Each fiber is then surrounded by plastic sheath

Benefits of Fiber Optics



- The bandwidth of the signal produced by the transmitting antenna is more important than the medium in determining transmission characteristics.
- **Optical Fiber – Benefits**
- greater capacity
 - data rates of hundreds of Gbps
- smaller size & weight
- lower attenuation
- electromagnetic isolation
- greater repeater spacing
 - 10s of km at least



Distinguish optical fiber from twisted pair or coaxial cable:



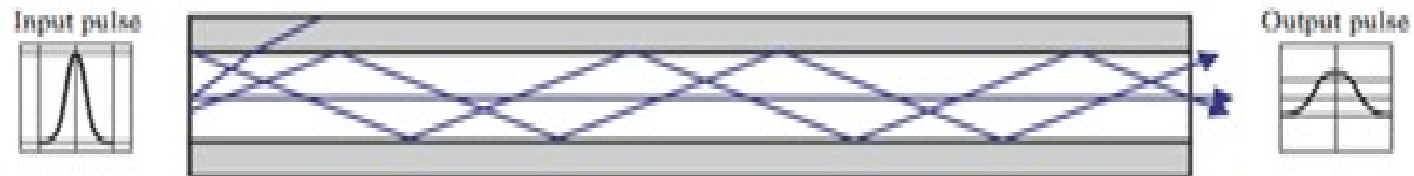
- ✓ **Greater capacity:** The potential bandwidth, and hence data rate, of optical fiber is immense; data rates of hundreds of Gbps over tens of kilometers have been demonstrated.
- ✓ **Smaller size and lighter weight:** Optical fibers are considerably thinner than coaxial cable or bundled twisted-pair cable.
- ✓ **Lower attenuation:** Attenuation is significantly lower for optical fiber than for coaxial cable or twisted pair, and is constant over a wide range.
- ✓ **Electromagnetic isolation:** Optical fiber systems are not affected by external electromagnetic fields. Thus the system is not vulnerable to interference, impulse noise, or crosstalk.
- ✓ **Greater repeater spacing:** Fewer repeaters mean lower cost and fewer sources of error.

Types of Optic fibre

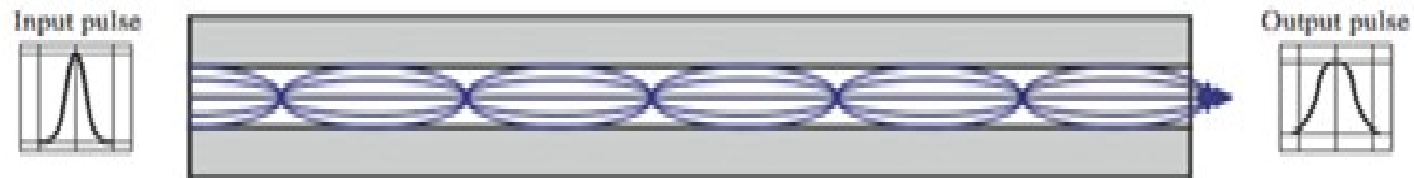


- Rays at shallow angles are reflected and propagated along the fiber; other rays are absorbed by the surrounding material. This form of propagation is called **step-index multimode**
- When the fiber core radius is reduced, fewer angles will reflect. By reducing the radius of the core to the order of a wavelength, only a single angle or mode can pass: the axial ray. This is **single-mode** propagation
- by varying the index of refraction of the core, a third type of transmission, known as **graded-index multimode**, is possible

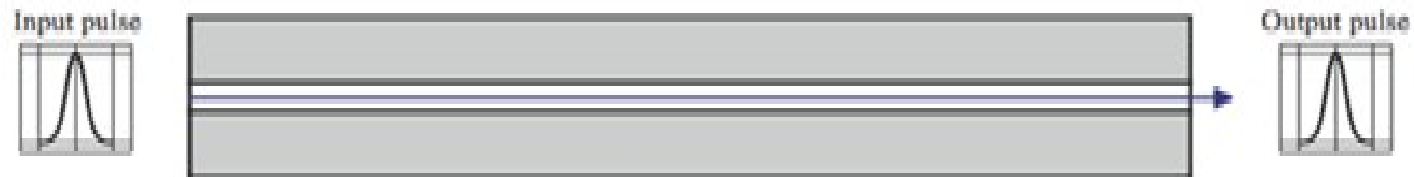
Optical Fibre Transmission Mode



(a) Step-index multimode



(b) Graded-index multimode



(c) Single mode

Optical Fiber - Transmission Characteristics

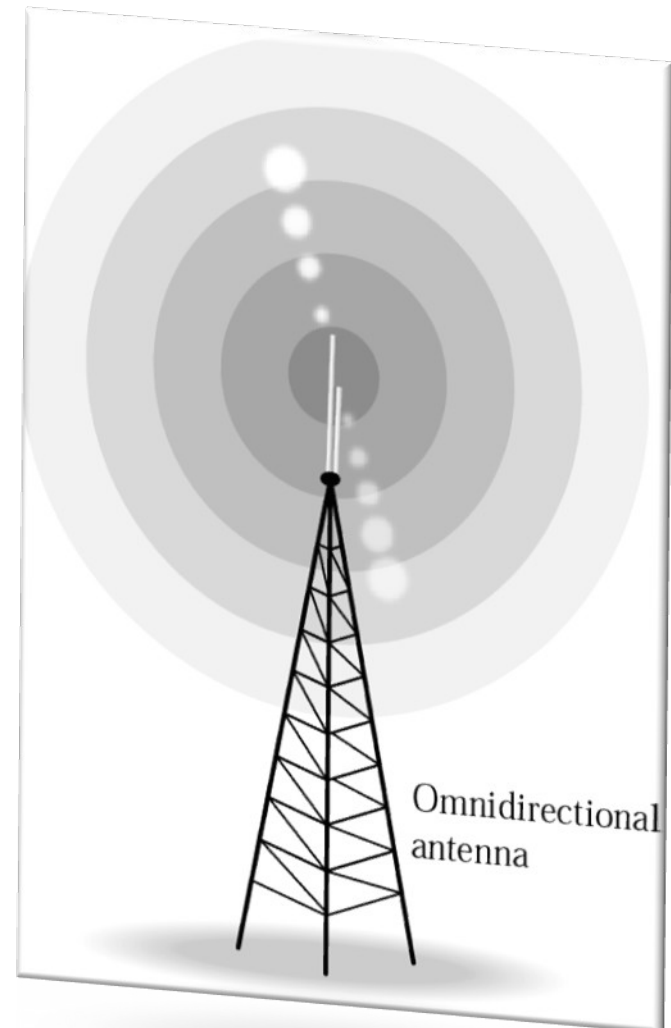


- uses total internal reflection to transmit light
 - effectively acts as wave guide for 10^{14} to 10^{15} Hz
- can use several different light sources
 - Light Emitting Diode (LED)
 - cheaper, wider operating temp range, lasts longer
 - Injection Laser Diode (ILD)
 - more efficient, has greater data rate
- relation of wavelength, type & data rate

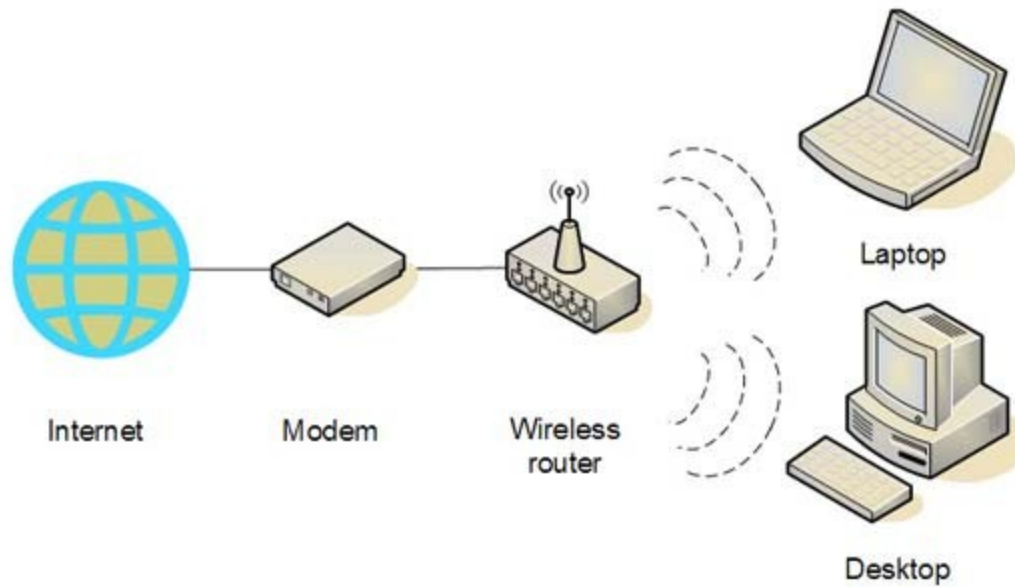
Wireless Transmission Frequencies



- 2GHz to 40GHz
 - microwave
 - highly directional
 - point to point
 - Satellite
- 30MHz to 1GHz
 - Omni directional
 - broadcast radio
- 3×10^{11} to 2×10^{14}
 - infrared
 - local



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Bluetooth



“Bluetooth is a low-cost, low power, short range wireless communication technology used in. networking, mobile phones and other portable device. Bluetooth wireless technology also enables devices to communicate with each other as soon as-they come within range; no need to connect, plug into, install, enable or configure anything.”



What Bluetooth Does Best



- Imagine being able to connect to the Internet on a dial-up connection you access through your mobile phone. Surfing the Internet then becomes possible anywhere your mobile phone can connect to your internet service provider.
- Perhaps you prefer to use a traditional mouse with your laptop. Choose a Bluetooth-enabled mouse and you won't have to keep track of a mouse cable.
- If you have a Bluetooth-enabled mobile phone that stores your business information in the Vcard format, you can easily share this information with your colleagues. Swap your Vcard with theirs, by wirelessly connecting to their Bluetooth-enabled mobile phones.

Bluetooth



Bluetooth printer module



PDA Adapter



Headsets



Bluetooth phones



GPS

Wi-Fi



“WiFi or Wireless Fidelity refers to the technology surrounding the radio transmission of internet protocol data from an internet connection wirelessly to a host computer. Most often the internet connection is a higher speed one such as satellite, DSL or cable rather than slower dial-up connections.”

► Home/SOHO network



A Wi-Fi network can be easily set up in a home or office and allow all users to share files, printers, and a single Internet connection. The network can be expanded by adding Wi-Fi radio cards to new devices, eliminating the need to purchase or install costly cabling.

How a Wi-Fi Network Works



- Basic concept is same as Walkie talkies.
- A Wi-Fi hotspot is created by installing an access point to an internet connection.
- An access point acts as a base station.
- When Wi-Fi enabled device encounters a hotspot the device can then connect to that network wirelessly.
- A single access point can support up to 30 users and can function within a range of 100 – 150 feet indoors and up to 300 feet outdoors.
- Many access points can be connected to each other via Ethernet cables to create a single large network.

Wi-Fi Applications



- Home
- Small Businesses or SOHO
- Large Corporations & Campuses
- Health Care
- Wireless ISP (WISP)
- Travellers

Wi-Fi Gadgets



AIBO talks 802.11



Laptop adapter



Wireless camera



Tablet PCs



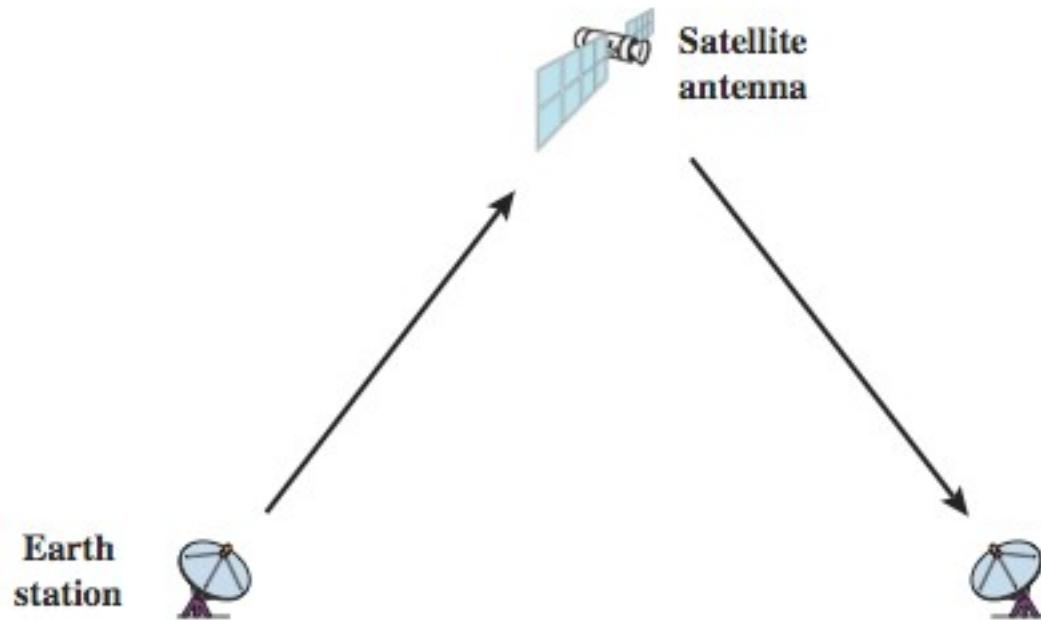
Access Point

Satellite Microwave



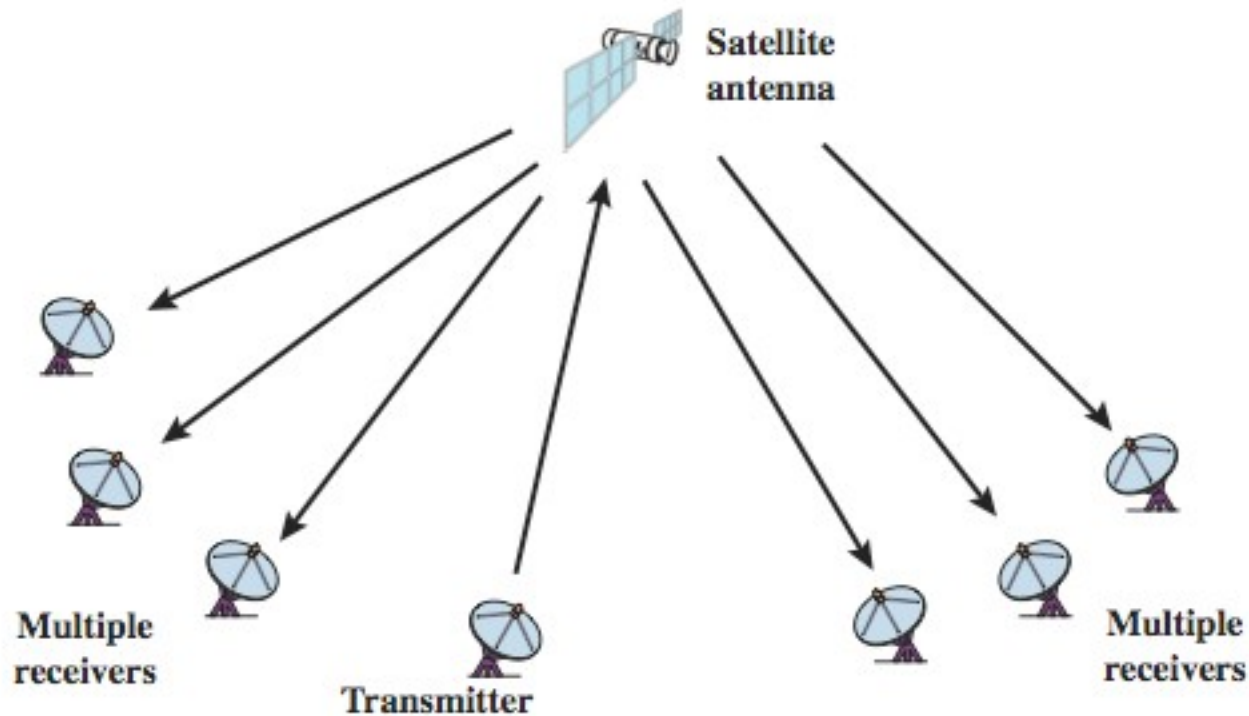
- Satellite is relay station
- Receives on one frequency, amplifies or repeats signal and transmits on another frequency
 - Eg. Uplink 5.925-6.425 ghz & downlink 3.7-4.2 ghz
- Typically requires geo-stationary orbit
 - Height of 35,784km
 - Spaced at least 3-4° apart
- Typical uses
 - Television
 - Long distance telephone
 - Private business networks
 - Global positioning

Satellite Point to Point Link



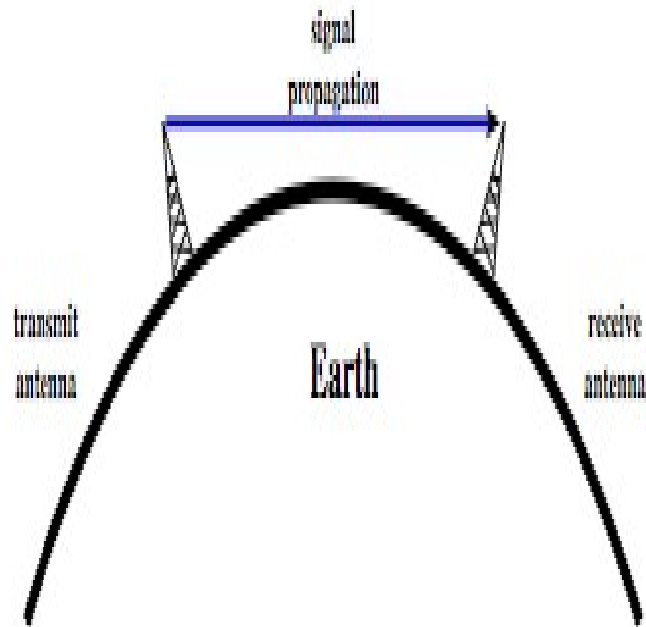
(a) Point-to-point link

Satellite Broadcast Link

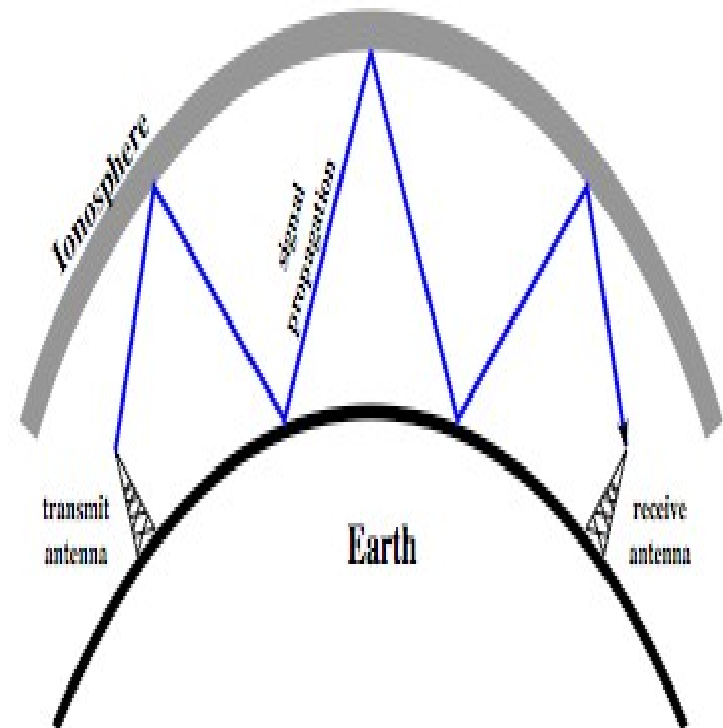


(b) Broadcast link

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(c) Line-of-sight (LOS) propagation (above 30 MHz)



(b) Sky-wave propagation (2 to 30 MHz)

Infrared



“Infrared technology allows computing devices to communicate via short-range wireless signals. With infrared, computers can transfer files and other digital data bidirectionally. The infrared transmission technology used in computers is similar to that used in consumer product remote control units.”

- Modulate noncoherent infrared light
- End line of sight (or reflection)
- Are blocked by walls
- No licenses required
- Typical uses
 - TV remote control
 - IRD port

Infrared



Installation and Usage - Computer infrared [network adapters](#) both transmit and receive data through ports on the rear or side of a device. Infrared adapters are installed in many laptops and handheld personal devices

Range - Infrared communications span very short distances. Place two infrared devices within a few feet (no more than 5 meters) of each other when networking them. Unlike Wi-Fi and Bluetooth technologies, infrared network signal cannot penetrate walls or other obstructions and work only in the direct "line of sight."

Performance - Infrared technology used in local networks exists in three different forms:

IrDA-SIR (slow speed) infrared supporting data rates up to 115 Kbps

IrDA-MIR (medium speed) infrared supporting data rates up to 1.15 Mbps

IrDA-FIR (fast speed) infrared supporting data rates up to 4 Mbps

IR Advantages:



- Low power requirements: therefore ideal for laptops, telephones, personal digital assistants
- Low circuitry costs: \$2-\$5 for the entire coding/decoding circuitry
- Simple circuitry: no special or proprietary hardware is required, can be incorporated into the integrated circuit of a product
- Higher security: directionality of the beam helps ensure that data isn't leaked or spilled to nearby devices as it's transmitted
- Portable

IR Disadvantages:



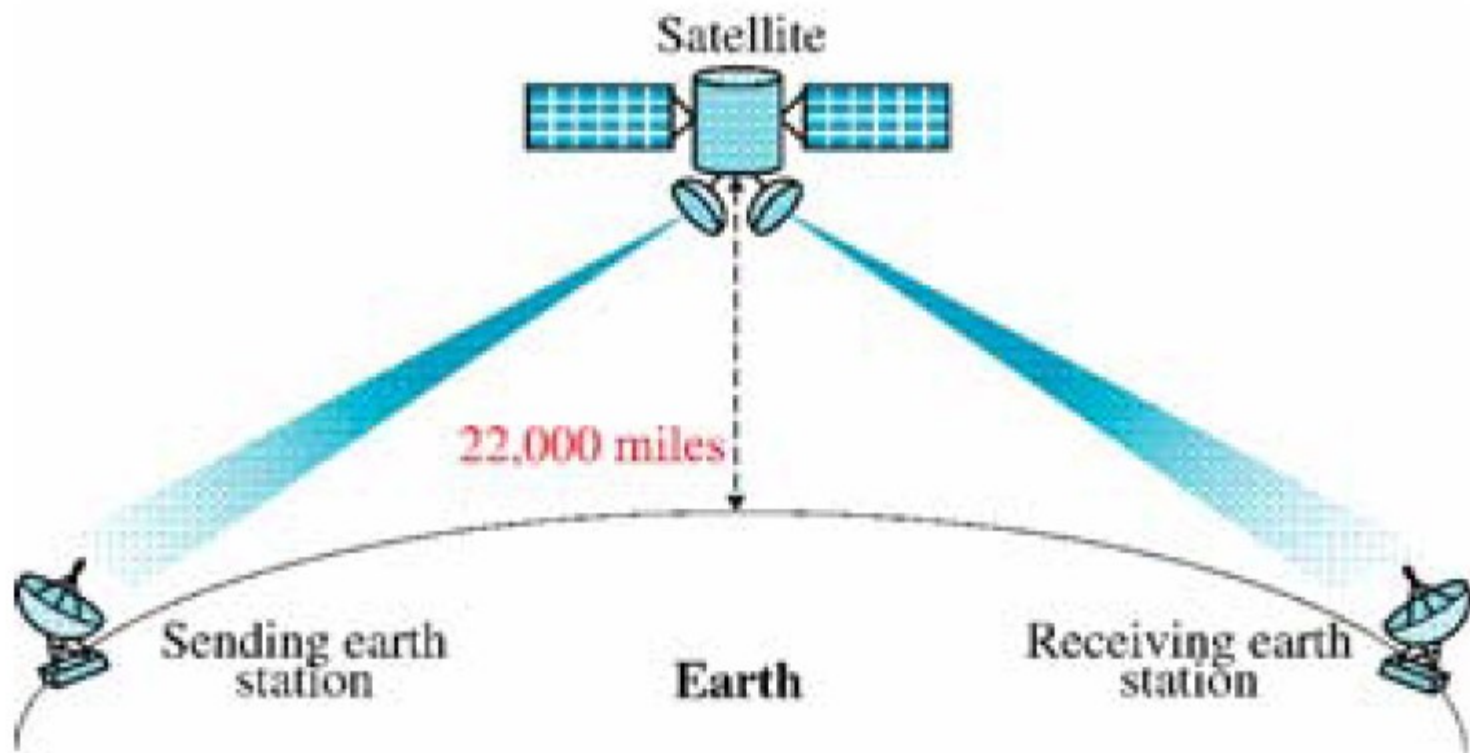
- Line of sight: transmitters and receivers must be almost directly aligned (i.e. able to see each other) to communicate
- Blocked by common materials: people, walls, plants, etc. can block transmission
- Short range: performance drops off with longer distances
- Light, weather sensitive: direct sunlight, rain, fog, dust, pollution can affect transmission
- Speed: data rate transmission is lower than typical wired transmission

Comparisiom of Major Wireless Transmission Media



Medium	Description and Advantages	Limitations and Drawbacks
Microwave	Point-to-point communication in lineof-sight path.	Must have visual contact between antennas.
	Antennas are used (30 miles apart).	Susceptible to environmental interferences
	Provide large capacity.	
	Can be done quickly at lower cost	
Satellites	Can be at high, medium, or low orbit; used in GPSs.	Expensive to build and maintain.
	Complete global coverage is available with three satellites.	
Radio/electromagnatic	Effective for short ranges; used in LANs	Limited range; difficult to secure.
	Inexpensive and easy to install.	Can create interference with communication devices.

Applications



A Wireless Connected home

